॥ श्री गणेशाय नमः ॥

Hypothesis Testing

A hypothesis test is a formal way to make a decision based on statistical analysis.

A hypothesis test has the following general steps:
1. Set up 2 contradictory hypotheses. One represents our assumption, the null hypothesis, H_0. The other is the alternate hypothesis, H_1 (or H_a).
2. Perform an experiment to collect data. Analyze the data using an appropriate distribution.
3. Decide whether the experimental data contradicts the hypothesis or not.
4. Translate the decision into a clear, non-technical conclusion.

Hypothesis tests are tests about the population parameters.

The null hypothesis (H_0) is a statement involving equality (<, >, =) about a population parameter.
We assume that the null hypothesis is true to do our analysis.
The alternate hypothesis (H_1) is a statement that contradicts the alternate hypothesis.
These generally involve inequalities (≠, >, <).

E.g. $F_\theta \sim N(\theta, 1)$

$H_0: \theta = 1, \quad H_1: \theta \neq 1$ → simple hypothesis: gives a fixed value to the population parameter

$H_0: \theta \leq 1, \quad H_1: \theta > 1$ → composite hypothesis: gives a range for the population parameter

A test for H_0 can be specified by defining a region C in the n-dim space such that the hypothesis is rejected if the random samples $X_1, \ldots, X_n$ lie in C and accepted otherwise.
The region C is called the critical region.

We accept H_0 if $X_1, \ldots, X_n$ do not lie in C.

Outcomes of the test and types of error:

Type-1 error: We decide to reject the null hypothesis when it is actually true.
Type-2 error: We decide to accept the null hypothesis when it is actually false.

Table of error types		Null hypothesis (H_0) is	
		True	**False**
Decision about null hypothesis (H_0)	**Fail to reject**	Correct inference (true negative) (probability = $1-\alpha$)	Type II error (false negative) (probability = β)
	Reject	Type I error (false positive) (probability = α)	Correct inference (true positive) (probability = $1-\beta$)

α is called the level of significance of the test.
1-β is called the power of the test.

α : level of significance = P(type-1 error) = P(hypothesis is rejected | hypothesis is true)
A level of significance 10%, i.e. α=0.10 means that the test will reject a correct hypothesis only in 10% of the cases.

Tests concerning the mean of the normal population

Case 1: Known variance - Z test

suppose that $X_1, \ldots, X_n$ is a random sample of size n from a normal distribution with an unknown mean and a known variance.

$$F_\mu : N(\mu, \sigma^2)$$

(μ: unknown; σ^2: known)

and suppose we want to test a null hypothesis $H_0 : \mu = \mu_0$
against the alt. hypothesis $H_1 : \mu \neq \mu_0$

Point estimator for $\mu = \bar{x}$, so $\bar{x}$ shouldn't be too far from μ_0 (the hypothesized mean value).

$$|\bar{x} - \mu_0| > c \longrightarrow \text{reject}$$

Since $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ is a P.E. for the mean, we create a test s.t. the critical region is given by

$$C = \{ x_1, \ldots, x_n \mid |\bar{x} - \mu_0| > c \} \quad \text{for a suitably chosen } c.$$

We desire a level of significance α, i.e. $P\{\text{type-1}\}$ error is at most α.

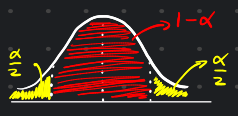
$$P\{ |\bar{x} - \mu_0| > c \} = \alpha$$

When $\mu = \mu_0$, $\bar{x} \sim N(\mu_0, \frac{\sigma^2}{n})$

$$\Rightarrow \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$P\{ |\bar{x} - \mu_0| > c \} = \alpha$$

$$\Rightarrow P\left\{ \frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} > \frac{c}{\sigma/\sqrt{n}} \right\} = \alpha$$



$$\Rightarrow P\left\{ |z| > \frac{c}{\sigma/\sqrt{n}} \right\} = \alpha$$

$$\Rightarrow 2 \cdot P\left\{ z > \frac{c}{\sigma/\sqrt{n}} \right\} = \alpha$$

$$\Rightarrow P\left\{ z > \frac{c}{\sigma/\sqrt{n}} \right\} = \alpha/2$$

We know that $P\{ z > z_\alpha \} = \alpha$
or $P\{ z > z_{\frac{\alpha}{2}} \} = \alpha/2$

$$\therefore \frac{c}{\sigma/\sqrt{n}} = z_{\alpha/2}$$

$$c = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The significance level α test to reject H_0 is $|\bar{x} - \mu_0| > z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$.

or we reject H_0 if $\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} > z_{\frac{\alpha}{2}}$

This is the z-test.

Q. Population ~ N(μ, 4). $H_0 : \mu = 8$, $H_1 : \mu \neq 8$

A sample of 5 items is taken from the population. $\bar{x} = 9.5$
Test the hypothesis with level of significance 0.05.

→ $\sigma^2 = 4$, $\bar{x} = 9.5$

we reject if $\frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}} > z_{\frac{\alpha}{2}}$

$$\frac{x - \mu_0}{\sigma/\sqrt{n}} = \frac{(9.5 - 8)\sqrt{5}}{2} = 1.677$$

$$z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$$

$$\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} < z_{\frac{\alpha}{2}}$$

we accept the hypothesis.

$P\{z > z_\alpha\} = \alpha$
$\Rightarrow 1 - \phi(z_\alpha) = \alpha$
$\Rightarrow \phi(z_\alpha) = 1 - \alpha$

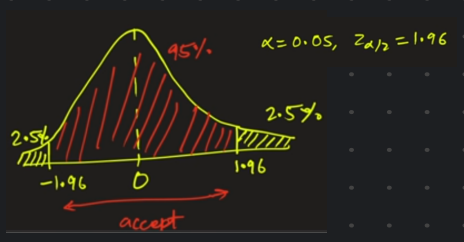
$\phi(z_{\alpha/2}) = 1 - \alpha/2$
$= 1 - 0.025$
$= 0.975$

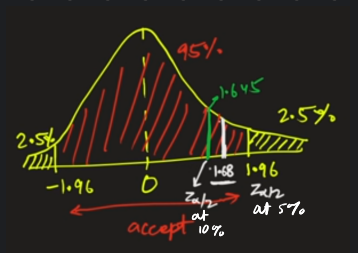
Q. Will we accept the hypothesis at l.o.s. = 0.1?

→ $\alpha = 0.1 \Rightarrow \frac{\alpha}{2} = 0.05$ $\quad z_{\frac{\alpha}{2}} = \varphi^{-1}(1 - \frac{\alpha}{2})$
$= \varphi^{-1}(0.95)$
$= 1.645$

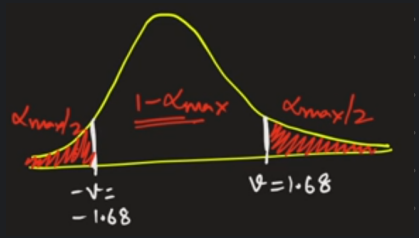
Since $\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} > z_{\frac{\alpha}{2}}$, we reject the hypothesis.

There's a maximum value of α at which we'll accept the hypothesis in a test.





The value $\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}}$ is called the test statistic.
↳ say v



$$\alpha_{max} = P\{|z| > v\}$$
$$= 2P\{z > v\}$$
$$= 2(1 - \varphi(v))$$

For above question, $\alpha_{max} = 2(1 - \varphi(1.68)) = 2(1 - 0.9535) = 0.093$
0.093 is called the p-value of the test.
For any $\alpha > 9.3\%$
or $\alpha > 0.093$, we'll reject the hypothesis.

Q. Population : $N(\mu, 4)$. $H_0 : \mu = 8$, $H_1 : \mu \neq 8$.
Sample : $n = 5$, $\bar{x} = 8.5$
Calculate the p-value for the test.

→ p-value = α_{max} for which test is accepted.

$$v = \frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} = \frac{8.5 - 8}{2/\sqrt{5}} = \frac{\sqrt{5}}{4} = 0.5590$$



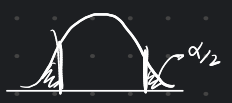
$$P\{z > 0.559\} = \frac{\alpha_{max}}{2}$$

$$\alpha_{max} = 2 \cdot P\{z > 0.5590\}$$
$$= 2(1 - \varphi(0.5590))$$
$$= 2(1 - 0.7123)$$
$$= 0.5754$$

$\alpha_{max} = 57.54\ \%$ $\quad 8 \approx 8.5$, so it'll accept for a lot of α values.

Q. what if $\bar{x} = 11.5$?

→
$$v = \frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}} = \frac{3.5}{2/\sqrt{5}} = 3.9131$$



$$\frac{\alpha_{max}}{2} = P\{z > 3.9131\}$$
$$= (1 - \varphi(3.9131))$$
$$= 1 - 0.99995$$
$$\frac{\alpha_{max}}{2} = 0.00005$$
$$\alpha_{max} = 0.0001$$

→ the hypothesis would be rejected in most of the cases.

Q. The average weight of all residents in a city is 168 lbs. A nutritionist believes that the true mean is different. She measures the weights of 36 individuals. The sample mean came out to be 169.5 lbs with a standard deviation of 3.9.
a. State the null and alternate hypotheses.
b. At 5% l.o.s., is the hypothesis accepted?
c. Calculate the p-value (the maximum l.o.s. at which the hypothesis will be accepted) for the z-test.

→ $H_0 : \mu = 168$

$H_1 : \mu \neq 168$

$\mu_0 = 168$, $\sigma = 3.9$, $\bar{x} = 169.5$, $n = 36$

b) $\alpha = 0.05$, $\alpha/2 = 0.025$, $z_{\alpha/2} = 1.96$

$$z\text{-statistic} = \frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} = \frac{169.5 - 168}{3.9/\sqrt{36}} = 2.3076$$

$$\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} > z_{\frac{\alpha}{2}} \quad \Rightarrow \text{Hypothesis is rejected.}$$

c) P-value :



$$\frac{\alpha}{2} = P\{z > 2.3076\}$$
$$\approx P\{z > 2.31\}$$
$$= 1 - \varphi(2.31)$$
$$\frac{\alpha}{2} = 1 - 0.9896$$
$$\alpha_{max} = 0.0208$$

→ for any $\alpha > 2.08\ \%$, H_0 will be rejected.

Type-2 error (β)

It is the probability of accepting a hypothesis when it isn't true.

$\beta(\mu)$

$= P\{\text{accepting } H_0 \mid H_0 \text{ is not true}\}$

$= P_\mu\left\{\frac{|\bar{X}-\mu_0|}{\sigma/\sqrt{n}} \le z_{\frac{\alpha}{2}}\right\}$ (condition of z-test)

But here the hypothesis is not true

$\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$ is not $N(0,1)$

So we sub. μ

$= P_\mu\left\{-z_{\frac{\alpha}{2}} \le \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\}$

$= P_\mu\left\{-z_{\frac{\alpha}{2}} - \frac{\mu}{\sigma/\sqrt{n}} \le \frac{\bar{x}-\mu_0-\mu}{\sigma/\sqrt{n}} \le z_{\frac{\alpha}{2}} - \frac{\mu}{\sigma/\sqrt{n}}\right\}$ ($\because z$)

$= P_\mu\left\{\left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right) - z_{\frac{\alpha}{2}} \le z \le z_{\frac{\alpha}{2}} + \left(\frac{\mu_0-\mu}{\sigma/\sqrt{n}}\right)\right\}$

$= \varphi\left[\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\frac{\alpha}{2}}\right] - \varphi\left[\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\frac{\alpha}{2}}\right]$

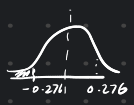
Q. Population ~ N(10, 4). $H_0 : \mu = 8$ (μ_0)
$H_1 : \mu \neq 8$

What is the probability that the z-test will be accepted for the null hypothesis (at l.o.s. 0.05).

$\beta = \varphi\left[\frac{\mu_0-\mu}{\sigma/\sqrt{n}} + z_{\frac{\alpha}{2}}\right] - \varphi\left[\frac{\mu_0-\mu}{\sigma/\sqrt{n}} - z_{\frac{\alpha}{2}}\right]$

$= \varphi\left[\frac{8-10}{2/\sqrt{5}} + z_{0.025}\right] - \varphi\left[\frac{8-10}{2/\sqrt{5}} - z_{0.025}\right]$

$= \varphi\left[-2.236 + 1.96\right] - \varphi\left[-2.236 - 1.96\right]$



$= \varphi[-0.276] - \varphi[-4.196]$

$= 1 - \varphi(0.276) - (1 - \underbrace{\varphi(4.196)}_{\approx 1})$

≈ 0

$= 1 - 0.6103$

$\beta \approx 0.39$

There's a 39% chance to accept the incorrect hypothesis.

Power function of the test:

The function $1-\beta(\mu)$ is called the power of the test. $\beta(\mu)$ is called the operating characteristic. Its graph is called the OC curve.

One-sided test

What if the set-up is such that the only alternative to $\mu=H_0$ is only $\mu>H_0$ (or $\mu<H_0$)?
So we're testing the hypotheses:

$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$

critical region: $\{(x_1, x_2, \dots, x_n) \in C\}$ then we reject the hypothesis.

$C: \{\bar{x} - \mu_0 > c\}$

we reject the hypothesis with atmost α prob.
↳ 0.05.

$P\{\bar{x} - \mu_0 > c\} = \alpha$ we know that $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$.

$P\left\{\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > \frac{c}{\sigma/\sqrt{n}}\right\} = \alpha$

$P\left\{z > \frac{c}{\sigma/\sqrt{n}}\right\} = \alpha$

$\frac{c}{\sigma/\sqrt{n}} = z_\alpha$

$c = z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

we accept H_0 if $\bar{x} - \mu_0 > c$
i.e. $\bar{x} - \mu_0 > z_\alpha \cdot \frac{\sigma}{\sqrt{n}}$

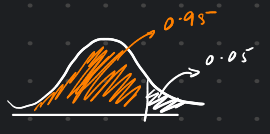
or $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > z_\alpha$ → accept. $\qquad \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} < z_\alpha$ → reject.

Q. Population ~ N(μ, 4). $H_0: \mu = 8$, $H_1: \mu > 8$

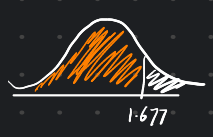
A sample of 5 items is taken from the population. $\bar{x} = 9.5$
Test the hypothesis with level of significance 0.05.

→ $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} = \frac{9.5-8}{2/\sqrt{5}} = 1.677$

$z_\alpha = z_{0.05}$
$= \varphi^{-1}(0.95)$
$= 1.645$



$\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > z_\alpha$ → rejected



$\alpha_{max} = P\{z > v\}$ (v: test-statistic)

$\alpha_{max} = 1 - \varphi(v) =$ p-value

here, p-value $= 1 - \varphi(1.68)$
$= 1 - 0.9535 = 0.0465$

$\alpha_{max} = 0.0465$ or 4.65%

so hypothesis will be accepted at any $\alpha < 4.65\%$
here $\alpha = 5\%$, so rejected.

$P(\text{type-2 error}) = \beta(\mu)$

$= P_\mu\{\text{accept } H_0 \mid H_0 \text{ is false}\}$

$= P_\mu\left\{\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} < z_\alpha\right\}$

$= P_\mu\left\{\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} < \frac{z_\alpha + \mu_0 - \mu}{\sigma/\sqrt{n}}\right\}$

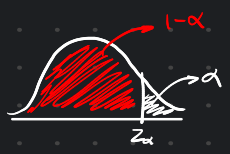
$= P_\mu\left\{z < \frac{z_\alpha + \mu_0 - \mu}{\sigma/\sqrt{n}}\right\}$

$= \varphi\left[z_\alpha + \frac{\mu_0 - \mu}{\sigma/\sqrt{n}}\right]$

If $\mu \uparrow$, $\beta(\mu) \uparrow$

$\beta(\mu_0) = \varphi(z_\alpha) = 1-\alpha$ —— ①

$1-\beta(\mu_0) = \alpha$



Axiom: The test $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} < z_\alpha$ for acceptance was designed for $H_0: \mu = \mu_0$, $H_1: \mu > \mu_0$

The same test works for $H_0: \mu \le \mu_0$, $H_1: \mu > \mu_0$.

Proof:

↳ The prob. of rejection should be $\le \alpha$ → we'll prove that for that test, this is still true for the new hypothesis H_0.

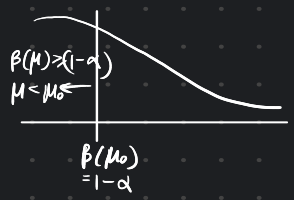
To prove: $1-\beta(\mu) \le \alpha \quad \forall \mu \le \mu_0$

or $\beta(\mu) \ge 1-\alpha \quad \forall \mu \le \mu_0$.

since $\beta(\mu) \uparrow$ as $\mu \downarrow$,

using eqn ①,

$\beta(\mu) \ge \beta(\mu_0) \quad \forall \mu \le \mu_0$



so the given hypothesis will work for

$H_0: \mu \le \mu_0 \quad H_1: \mu > \mu_0$.

We can also test the other one sided hypothesis.

$H_0: \mu = \mu_0$ (or $\mu \ge \mu_0$) $\quad H_1: \mu_0 < \mu \quad$ at l.o.s. α

we accept the hypothesis if $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > -z_\alpha$ (similar to one-sided C.I.)

and reject it otherwise.

Critical region:

$C: \{\bar{x}-\mu_0 > c\}$

we reject the hypothesis with at most probability α.

$\therefore$ (previous one) $P\{\bar{x}-\mu_0 > c\} = \alpha$

$\Rightarrow P\left\{\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > \frac{c}{\sigma/\sqrt{n}}\right\} = \alpha$

$\Rightarrow P\left\{z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$

$1-\frac{c\sqrt{n}}{\sigma} = z_\alpha$

$\frac{c\sqrt{n}}{\sigma} = 1-z_\alpha$

$\frac{c\sqrt{n}}{\sigma} = -z_\alpha \Rightarrow c = \frac{-z_\alpha \sigma}{\sqrt{n}}$



For other side → $\bar{x}-\mu_0 < c$ ②

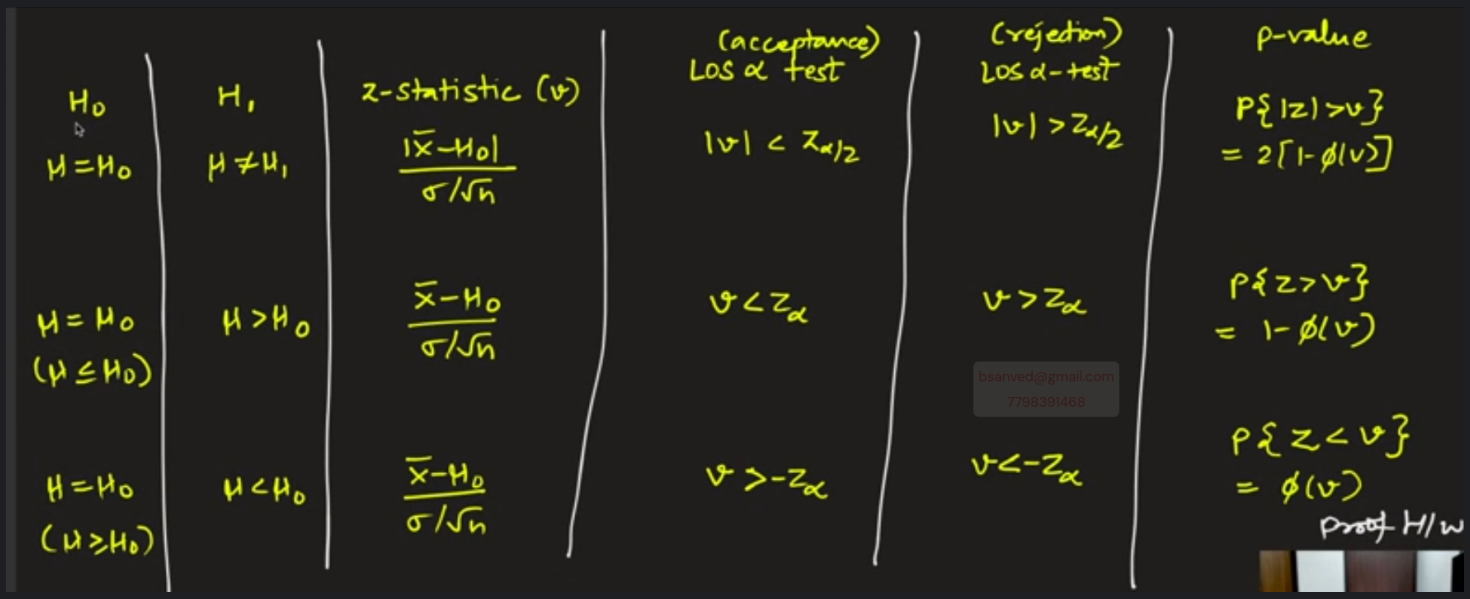
we know $\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim z$

$P\{\bar{x}-\mu_0 < c\} = \alpha$

$P\left\{z < \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$

$P\left\{z > 1-\frac{c\sqrt{n}}{\sigma}\right\}$

$\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} > -z_\alpha$ → accept



columns:
mu_0 mu_1 z-statistic los alpha test(acceptance) los alpha test(rejection) p-value

rows:

mu = mu_0 mu \not= mu0 (|xbar - mu_0|)/(sigma/sqrt n) |v| < z_(alpha/2) |v| < z_(alpha/2)

Case 1: Unknown variance - t test

$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$

We want $P\{H_0 \text{ is rejected} \mid H_0 \text{ is true}\} \leq \alpha$

Critical region: $P\{(x_1, \ldots, x_n) \in C\}$ then we reject the hypo.

$$C: P\{|\bar{x} - \mu_0| > c\} = \alpha$$

$$\bar{x} \sim N(\mu, \sigma^2/n) \quad (\sigma^2 \rightarrow \text{unknown})$$

$$\bar{x} = \frac{x_1 + \ldots + x_n}{n} \qquad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

We know that

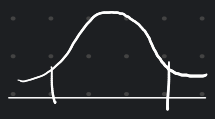
$$\frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} = T$$

$$P\left\{\frac{|\bar{x} - \mu_0|}{s/\sqrt{n}} > \frac{c}{s/\sqrt{n}}\right\} = \alpha$$

$$P\left\{|T| > \frac{c}{s/\sqrt{n}}\right\} = \alpha$$

$$2 \cdot P\left\{T > \frac{c}{s/\sqrt{n}}\right\} = \alpha$$

$$P\left\{T > \frac{c}{s/\sqrt{n}}\right\} = \frac{\alpha}{2}$$



We know that: $P\{T > t_{\alpha, n-1}\} = \alpha$

$$\therefore \frac{c}{s/\sqrt{n}} = t_{\frac{\alpha}{2}, n-1}$$

$$c = t_{\frac{\alpha}{2}, n-1} \cdot \frac{s}{\sqrt{n}}$$

$\therefore$ We reject the hypothesis if $|\bar{x} - \mu_0| > t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$

or

We accept H_0 if $\frac{|\bar{x} - \mu_0|}{s/\sqrt{n}} < t_{\frac{\alpha}{2}, n-1}$

Q. A public health official claims that the mean home water usage is 350 gallons / day. To verify this, a study of 20 randomly selected houses was done and the sample mean and standard deviation came out to be 353.8 and 21.85 gallons respectively. Is the claim accepted at 5% l.o.s.?

$\rightarrow$ $H_0 : \mu = 350$

$H_1 : \mu \neq 350$

$\mu_0 = 350, \ \bar{x} = 353.8, \ s = 21.85, \ n = 20.$

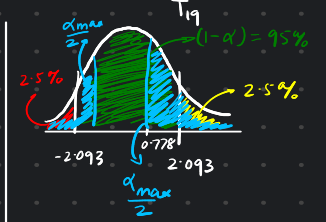
$$\frac{|\bar{x} - \mu_0|}{\sigma/\sqrt{n}} = \frac{353.8 - 350}{21.85/\sqrt{20}} = \frac{(3.8)\sqrt{20}}{21.85} = 0.7777$$

(t-statistic)

$$t_{\frac{\alpha}{2}, n-1} = t_{0.025, 19} = 2.093$$

t-statistic $< t_{\frac{\alpha}{2}, n-1}$

$\therefore$ The hypothesis is accepted.



$$\alpha_{max} = P\{|T_{19}| > 0.778\}$$

$$= 2 \cdot P\{T_{19} > 0.778\}$$

(can't get this from the t-table)

One-sided t-test:

①

$H_0: \mu = \mu_0$ (or $\mu \le \mu_0$) $\quad H_1: \mu > \mu_0$

t-statistic: $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ → NOTE: no absolute value

Acceptance: $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}} < t_{\alpha, n-1}$

②

$H_0: \mu = \mu_0$ (or $\mu \ge \mu_0$) $\quad H_1: \mu < \mu_0$

Acceptance: $\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}} > -t_{\alpha, n-1}$

Q. The manufacturer of a new fibre-glass tyre claims that the average life will be at least 40,000 miles. To verify this, a sample of 12 tyres is tested and the sample mean and standard deviation came out to be 37283 and 2732 miles respectively.
Test the manufacturer's claim at 5% l.o.s.

→ $H_0: \mu \ge 40{,}000$

$H_1: \mu < 40{,}000$

$n = 12, \quad \alpha = 0.05$

$\bar{x} = 37283, \quad s = 2732$

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{37283 - 40000}{2732/\sqrt{12}} = -3.445$$

$t_{0.05, 11} = 1.796 \; ; \quad -t_{\alpha, n-1} = -1.796$

$T < -t_{0.05, 11}$

rejected.

H_0	H_1	t-stastic v	accept LOS α test	reject LOS α test	p-value α_{max}
$\mu = \mu_0$	$\mu \ne \mu_0$	$\dfrac{\lvert\bar{x} - \mu_0\rvert}{s/\sqrt{n}}$	$v < t_{\alpha/2, n-1}$	$v > t_{\alpha/2, n-1}$	$2 \cdot P\{T_{n-1} > v\}$
$\mu \le \mu_0$	$\mu > \mu_0$	$\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$v < t_{\alpha, n-1}$	$v > t_{\alpha, n-1}$	$P\{T_{n-1} > v\}$
$\mu \ge \mu_0$	$\mu < \mu_0$	$\dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$v > -t_{\alpha, n-1}$	$v < -t_{\alpha, n-1}$	$P\{T_{n-1} < v\}$



Testing 2 normal populations

Case 1: Known variances

$X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_m$ are independent samples from 2 normal distributions $X(\mu_x, \sigma_x^2)$ and $Y(\mu_y, \sigma_y^2)$ resp.
μ_x and μ_y are unknown.
σ_x^2 and σ_y^2 are known.

$H_0: \mu_x = \mu_y$ or $\mu_x - \mu_y = 0$
$H_1: \mu_x \neq \mu_y$ or $\mu_x - \mu_y \neq 0$

we define a critical region $\{(x_1, \dots, x_n, y_1, \dots, y_n) \in C\}$ where we reject H_0.

$$C: \{|\bar{x} - \bar{y}| > c\} \qquad \bar{x} = \frac{x_1 + \dots + x_n}{n}, \quad \bar{y} = \frac{y_1 + \dots + y_n}{m}$$

$$P\{|\bar{x} - \bar{y}| > c\} = \alpha$$

we know that $\bar{x} - \bar{y} \sim \left(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$

$$\frac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

$$P\left\{\frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} > \frac{c}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right\} = \alpha \qquad \text{given } \mu_x = \mu_y$$

$$z\text{-statistic} = \text{pivot} = \frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$$

$$P\left\{|z| > \frac{c}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right\} = \alpha$$

$$2 \cdot P\left\{z > \frac{c}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right\} = \alpha$$

$$P\left\{z > \frac{c}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right\} = \frac{\alpha}{2}$$

we know that $P\{z > z_{\frac{\alpha}{2}}\} = \alpha/2$

$$\therefore \frac{c}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} = z_{\frac{\alpha}{2}}$$

$$c = z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$$

So we reject H_0 if $|\bar{x} - \bar{y}| > z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$

$$\text{or } \frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} > z_{\frac{\alpha}{2}}$$

Q. You are testing 2 cold medicines M1 and M2. There are 22 volunteers of which 10 (group X) are given M1 and 12 (group Y) are given M2.
The sample mean times taken to treat cold on these people came out to be $\bar{X} = 6.45$, $\bar{Y} = 7.125$.
Assume that the population variances are $\sigma_x^2 = 0.581$, $\sigma_y^2 = 0.778$.
A claim says that the average times taken by M1 and M2 are equal. Test the hypothesis at 5% l.o.s.

$$\rightarrow \quad \frac{|\bar{x}-\bar{y}|}{\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}} = \frac{|6.45-7.125|}{\sqrt{\frac{0.581}{10}+\frac{0.778}{12}}} = 1.925$$

$$z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$$

$$z < z_{\frac{\alpha}{2}} \quad \therefore \text{ we accept } H_0, \text{ i.e. } \mu_x = \mu_y \text{ at 5\% l.o.s.}$$

$$\alpha_{max} = P\{|z| > v\}$$
$$= 2 \cdot P\{z > v\}$$

p-value $= 2(1-\Phi(v))$

One-sided test

$H_0 : \mu_x = \mu_y \qquad H_1 : \mu_x > \mu_y$
$(\text{or } \mu_x \leq \mu_y)$

we accept H_0 if $\dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}} < z_\alpha$

$H_0 : \mu_x = \mu_y \qquad H_1 : \mu_x < \mu_y$
$(\text{or } \mu_x \geq \mu_y)$

we accept H_0 if $\dfrac{\bar{x}-\bar{y}}{\sqrt{\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}}} > -z_\alpha$

Case 2: Unknown and equal variances

$X_1, X_2, \dots, X_n \sim N(\mu_x, \sigma^2)$
$Y_1, \dots, Y_n \sim N(\mu_y, \sigma^2)$

μ_x, μ_y are unknown.
σ is unknown (we assume both variances are equal $= \sigma$).

$H_0: \mu_x = \mu_y$ $\quad$ $H_1: \mu_x \neq \mu_y$

$s_p^2 = \dfrac{(n-1)s_x^2 + (m-1)s_y^2}{n+m-2}$ ⟶ weighted avg.

We know that $\dfrac{\bar{x} - \bar{y} - (\mu_x - \mu_y)}{s_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t_{n+m-2}$

$P\{|\bar{x} - \bar{y}| > c\} = \alpha$

$P\left\{\dfrac{|\bar{x} - \bar{y}|}{s_p\sqrt{\frac{1}{n} + \frac{1}{m}}} > c\right\} = \alpha$

t-statistic (v) $= \dfrac{|\bar{x} - \bar{y}|}{\sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}}$

We accept H_0 if $v < t_{\frac{\alpha}{2}, n-m+2}$

p-value: $P\{|T_{n+m-2}| > v\}$

$= 2 \cdot P\{T_{n+m-2} > v\}$

One-sided test

$H_0: \mu_x = \mu_y$ (or $\mu_x \leq \mu_y$) $\quad$ $H_1: \mu_x > \mu_y$

We accept H_0 if $\dfrac{\bar{x} - \bar{y}}{\sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}} < t_{\alpha, n+m-2}$

$H_0: \mu_x = \mu_y$ (or $\mu_x \geq \mu_y$) $\quad$ $H_1: \mu_x < \mu_y$

We accept H_0 if $\dfrac{\bar{x} - \bar{y}}{\sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}} > -t_{\alpha, n+m-2}$

Eg: Testing 2 cold medicines M1 and M2.

22 volunteers for a test, (x) 10 are given M1, (y) 12 are given M2. Time taken to treat cold is tested on these people and the sample mean came out to be $\bar{x} = 6.45$, $\bar{y} = 7.125$, and the sample variance $S_x^2 = 0.581$, $S_y^2 = 0.778$, also assume the pop. variance equal

Claim: M_2 reduces the mean length cold lasts then taking M_1. Test the hypothesis at 5% level of significance.

→

$\bar{x} = 6.45$ $\quad \bar{y} = 7.125$ $\quad \alpha = 5\%$

$S_x^2 = 0.581$ $\quad S_y^2 = 0.778$

$n = 10$ $\quad m = 12$

$H_0: \mu_x \geq \mu_y \quad H_1: \mu_x < \mu_y$

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2} = \frac{9(0.581) + 11(0.778)}{10+12-2} = 0.689$$

$$\frac{\bar{x} - \bar{y}}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} = \frac{6.45 - 7.125}{0.689\sqrt{\frac{1}{10} + \frac{1}{12}}} = -1.8987 \approx -1.90$$

$t_{\alpha, n+m-2} = t_{0.05, 20} = 1.725$; $\quad -t_{\alpha, n+m-2} = -1.725$

t-statistic $< -t_{\alpha, n+m-2}$

∴ we reject H_0 at l.o.s. 0.01.

Case 2: Unknown and unequal variances (approx. test)

$X_1, \dots, X_n \sim X(\mu_x, \sigma_x^2)$
$Y_1, \dots, Y_m \sim Y(\mu_y, \sigma_y^2)$
means and variances are unknown.

If n, m are large,
S_x and S_y can be used as approx. values of σ_x and σ_y.

$H_0 : \mu_x = \mu_y, \quad H_1 : \mu_x \neq \mu_y$

$$\text{Test-statistic } (v) = \frac{|\bar{x} - \bar{y}|}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}} \sim Z \quad (\text{if } n, m \text{ are large})$$

Then we accept H_0 if $v < z_{\frac{\alpha}{2}}$

This is an approximate test. It gets more accurate for larger sample sizes (n and m).

H_0	H_1	Test statistic (v)	Significance level α-test accept	reject	p-value
$\mu_x = \mu_y$ (σ_x, σ_y are known)	$\mu_x \neq \mu_y$	$\frac{\lvert\bar{x} - \bar{y}\rvert}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}$	$v < z_{\alpha/2}$	$v > z_{\alpha/2}$	$P\{\lvert Z\rvert > v\} = 2\{1 - \phi(v)\}$
$\mu_x = \mu_y$ (σ_x, σ_y are unknown but equal)	$\mu_x \neq \mu_y$	$\frac{\lvert\bar{x} - \bar{y}\rvert}{\sqrt{\frac{S_p^2}{n} + \frac{S_p^2}{m}}}$, $S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$	$v < t_{\alpha/2, n+m-2}$	$v > t_{\alpha/2, n+m-2}$	$P\{\lvert T_{n+m-2}\rvert > v\} = 2 \cdot P\{T_{n+m-2} > v\}$
$\mu_x = \mu_y$ (σ_x, σ_y are unknown but equal)	$\mu_x \neq \mu_y$	$\frac{\lvert\bar{x} - \bar{y}\rvert}{\sqrt{\frac{S_p^2}{n} + \frac{S_p^2}{m}}}$, $S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$	$v < t_{\alpha/2, n+m-2}$	$v > t_{\alpha/2, n+m-2}$	$P\{\lvert T_{n+m-2}\rvert > v\} = 2 \cdot P\{T_{n+m-2} > v\}$
$\mu_x = \mu_y$ (σ_x, σ_y are unknown, unequal, n, m are large)	$\mu_x \neq \mu_y$	$\frac{\lvert\bar{x} - \bar{y}\rvert}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}}$	$v < z_{\alpha/2}$	$v > z_{\alpha/2}$	$P\{\lvert Z\rvert > v\} = 2[1 - \phi(v)]$

Tests involving the variance of a normal population

Chi-squared test

Let $X_1, \ldots, X_n$ be a random sample from a normal population (μ, σ^2) (μ and σ^2 are unknown).
We want to test the hypothesis:

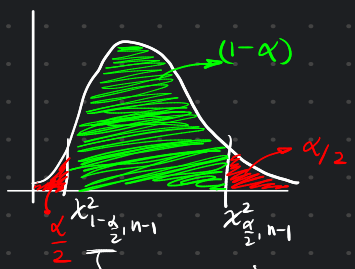
$H_0: \sigma^2 = \sigma_0^2 \qquad H_1: \sigma^2 \neq \sigma_0^2$

We know that $(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$

For our test,

$$P\left\{ \chi^2_{1-\frac{\alpha}{2},n-1} \leq (n-1)\frac{S^2}{\sigma_0^2} \leq \chi^2_{\frac{\alpha}{2},n-1} \right\} = 1-\alpha$$

(acceptance)



as curve is a symmetric, we can't write it in terms of the other

$$P\left\{(n-1)\frac{S^2}{\sigma_0^2} > \chi^2_{\frac{\alpha}{2},n-1}\right\} + P\left\{(n-1)\frac{S^2}{\sigma_0^2} < \chi^2_{1-\frac{\alpha}{2},n-1}\right\} = \alpha$$

We accept H_0 if $\chi^2_{1-\frac{\alpha}{2},n-1} \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\frac{\alpha}{2},n-1}$

One-sided test:

$H_0: \sigma^2 \leq \sigma_0^2 \qquad H_1: \sigma^2 > \sigma_0^2$

we accept H_0 if $\frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\alpha,n-1}$

$H_0: \sigma^2 \geq \sigma_0^2 \qquad H_1: \sigma^2 < \sigma_0^2$

we accept H_0 if $\frac{(n-1)S^2}{\sigma_0^2} \geq \chi^2_{1-\alpha,n-1}$

Q. A machine that automatically controls the amount of ribbon on a tape is installed. The claim is that the standard deviation σ of the amount of ribbon is less than 0.15cm. If a sample of 20 tapes yields a sample variance of $s^2 = 0.025\text{cm}^2$, then test the hypothesis at 5% l.o.s.

→ $H_0: \sigma < 0.15$ or $\sigma^2 < 0.0225 \qquad H_1: \sigma \geq 0.15$

$$(n-1)\frac{S^2}{\sigma_0^2} = (19)\cdot\frac{0.025}{0.0225} = 21.1111$$

$$\chi^2_{0.05,19} = 30.144$$

$$(n-1)\frac{S^2}{\sigma_0^2} < \chi^2_{0.05,19} \Rightarrow \text{we accept the hypothesis.}$$

α-max or p-value: $\sigma_{max} = P\left\{\chi^2_{\alpha,19} > 21.11\right\}$

Paired t-test

Q. An industrial safety program was recently installed in the computer-chip. The average weekly loss in labor hours due to accidents before and after the program are:

Plant	1	2	3	4	5	6	7	8	9	10
Before	30.5	18.5	24.5	32	16	15	23.5	25.5	28	18
After	23	21	22	28.5	14.5	15.5	24.5	21	23.5	16.5

} } > not independent

Determine at 5% l.o.s. whether the safety program was not effective.

→ $H_0: \mu_A - \mu_B \geq 0$ $\quad H_1: \mu_{A-B} < 0$

or $\mu_{A-B} \geq 0$

(After − Before) $-7.5 \quad 2.5 \quad -2.5 \quad -3.5 \quad -1.6 \quad 0.5 \quad 1 \quad -4.5 \quad -4.5 \quad -1.5$

$\bar{x} = 2.16$

$s^2 = 8.989 \quad \left(\frac{\Sigma (x_i - \bar{x})^2}{(n-1)}\right)$

$t\text{-statistic} = \frac{\bar{x} - \mu_0}{s^2/n} = -2.266 \qquad -t_{\alpha, n-1} = -t_{0.05, 9} = -1.833$

As $t\text{-statistic} < -t_{\alpha, n-1}$, H_0 is rejected.